Problem K. Sticks

Input file:	standard input
Output file:	standard output
Time limit:	2 seconds
Memory limit:	1024 megabytes

Little Cyan Fish has an $n \times n$ matrix. Each row and each column has a stick on its left side and top side, respectively. Let x_i represent the length of the stick on the left side of the i-th row and y_i represent the length of the stick on the top side of the i-th column, where $0 \le x_i, y_i \le n$ and both are integers. Additionally, the sticks must not intersect, meaning there should be **no** $i, j \in [1, n]$ such that both $x_i \ge j$ and $y_j \ge i$ hold true.

Little Cyan Fish defines the matrix A as follows:

- For each $i, j \in [1, n]$, if $x_i \ge j$ or $y_j \ge i$, then $A_{i,j} = 1$; otherwise, $A_{i,j} = 0$.

Given an $n \times n$ matrix M containing `0`s, `1`s, and `?`s, you need to determine how many different matrices can be formed by replacing each `?` with either a `0` or a `1`, so that there is at least one set of the sticks $\{x_1, x_2, \cdots, x_n, y_1, y_2, \cdots, y_n\}$ that could obtain this matrix. Since the answer may be large, output it modulo 998 244 353.

Input

The first line of the input contains a single integer n ($1 \le n \le 3\,000$).

The next n lines of the input describes the matrix M. Each of the line contains a string of length n containing "`0`", "`1`", and "`?`", indicating the matrix.

Output

Output a single line contains a single integer, indicating the answer.

Examples

standard input	standard output
2 ?? ??	14
5 ??1?? ?1??0 ??0?? ???1? ??1??	3144
10 0000000000 ?????????? ?????????? ?????????? ?????????? ?????????? ?????????? ?????????? ?????????? ??????????	361458985